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Submitted by email to [REDACTED]

Re: Request for Information (RFI) on the Development of an Artificial Intelligence (AI) Action Plan ("Plan")

INTRODUCTION

Fathom staunchly supports the work of the Office of Science and Technology Policy to define and prioritize policy actions required to strengthen America's position in AI. President Trump and his administration have a unique opportunity to enable American AI leadership – the following recommendations will support that vision and help make it a reality.

AI could empower humanity to discover new materials, develop life-saving medications, and harness novel energy sources—an industrial and scientific revolution. It could transform education, media, and communication—an information revolution. It could fundamentally reshape how we live, work, and interact with one another—a socioeconomic revolution. It could unlock achievements once thought impossible: deciphering ancient texts previously considered lost to time, advancing scientific and mathematical theory, and creating entirely new forms of artistic expression—a renaissance. This is the extraordinary potential, and perhaps one day the extraordinary legacy, of advanced AI.

We must face this transition with optimism, humility, flexibility, and proactivity. The choices we make now—about how we develop, deploy, and govern AI—will determine whether we harness its extraordinary potential or allow it to unfold haphazardly, or even not at all. To act prudently, we first need to know where we want to end up together as members of a free and varied society, as individuals, partners, parents, teachers, community members, business leaders, factory workers, artists, academics, church parishioners, and civil society advocates.

Over the past year, Fathom has convened everyday Americans, national security experts, AI developers, and more. We conducted multiple polls, spent hundreds of hours conversing with stakeholders from across society, and hosted The Ashby Workshops, which brought together over 180 experts from business, government, academia, and civil society.

These voices, so varied in their expertise and worldviews, all share a concern for the preservation and deepening of American values – that the United States, not China, must continue to lead in AI.

As one influential stakeholder put it, "we must all recommit to defending the principles that made us great," and brought us to this moment of great possibility. AI systems rely on vast datasets made possible by the internet—itsself built upon the ideals embedded in America's founding: the free exchange of ideas, speech, and information. The dynamic innovation ecosystem, substantial industrial infrastructure, and energy capacity required for AI development depend on robust protections for private enterprise, property rights, and free and fair competition. Likewise, vital safeguards for workers and consumers are critical to ensure that while business prospers, society remains confident that innovation will not trample our unalienable rights. By standing on these principles rather than abandoning them, and in some cases remembering them once again, we can navigate AI's complexities with a shared sense of purpose and in a way that unites rather than divides us, turning technological change and the resulting disruption into a catalyst that benefits all Americans.

Fathom's mission is to find, build, and scale the solutions needed for our transition to a world with AI. Our research indicates that a successful transition is one in which our shared values persist amid the upheavals we are sure to encounter. Therefore, Fathom recommends President Trump's AI Action Plan prioritize the protection and promotion of American values with, through, and despite AI.

Fathom's specific recommendations are structured around the following three insights:

1. **The U.S. must retain its global lead in AI to ensure the future of this technology and the societies it will shape are molded by Western values.** Leading means outpacing strategic competitors in both technical innovation and the widespread, trusted adoption of AI, ensuring that the future of this technology is shaped by our values, not those of our adversaries.
2. **AI should augment human potential, not replace it, while improving daily life, especially in critical areas like healthcare and public services.** Realizing this vision requires reimagining government and transforming science. It also requires us to commit to building systems whose code we understand and control, ensuring our technologies reflect our values.
3. **We need sensible rules of the road that foster innovation and security.** Neither government nor industry alone should dictate these rules; collaboration across government *and* industry is vital to support a vibrant start-up ecosystem and maintain our technological advantage. Realizing this vision requires governance innovation.

RECOMMENDATIONS

Based on our insights, Fathom respectfully recommends President Trump's AI Action Plan include the following priority actions:

Invest in Short- and Long-Term AI Governance

Like all general-purpose technologies, AI will present novel governance challenges. Sometimes, the optimal solution to these challenges will take the form of regulation. But often, they will require more imaginative thinking. To achieve effective AI governance, hasten the adoption of the technology across American society, and solidify our leadership position in the development

of this revolutionary technology, President Trump's AI Action Plan should consider the following:

1. Focus on Regulating Harmful Conduct: Successful regulation must focus on deterring harmful conduct. Of course, existing law already codifies countless behaviors that policymakers have desired to disincentivize through civil or criminal penalties – everything from lying on accounting forms to treason. Thus, in many ways the challenge lawmakers face is the application of existing law to the conduct of individuals and businesses using and developing AI. In many cases, the solution will be fairly straightforward; for example, fraud remains fraud regardless of whether it is performed using pen and paper or an advanced superintelligent agent. However, enforcing laws against people using advanced agents is likely to be a nontrivial task, requiring rapid government adoption of AI.

In other cases, AI may require correcting gaps in existing law, or even loosening law to enable productive uses inadvertently made unlawful through the design of existing regulatory structures. An example of the former is the knowing distribution of malicious deepfakes, which is not necessarily a crime in every US state. An example of the latter might be Food and Drug Administration clinical trials for new pharmaceuticals. The clinical trial system was designed on the assumption that drugs would be made for large populations of people with the same disease. But modern biology and AI are pushing toward a future where therapeutics are increasingly tailored to the specific medical needs of small groups of patients or even just a single patient. This could represent a paradigm shift in how we have historically viewed ultra-rare diseases and their treatment, and may require an entirely new form of drug approvals for treatments of this kind.

Regulating conduct does not mean that developers of frontier AI models should be free from regulations that apply to them specifically. Indeed, the development of AI is a novel kind of conduct that may well require novel forms of regulatory intervention.

2. Build an AI Evaluations Ecosystem: Even in an AI regulatory paradigm that relies primarily on existing law, businesses will face uncertainty about managing AI risks. Already, businesses across the country are struggling to evaluate models for reliability and for compliance with existing laws. AI policy thus often finds itself with a kind of chicken-and-egg problem: businesses do not know how to comply, and policymakers do not know how to describe what compliance looks like in a robust and measurable way.

The solution to the chicken-and-egg problem faced by regulators and businesses is a robust technical means of evaluating, auditing, and verifying the validity of models, training data, and model outputs.

One important challenge with evaluations, however, is that they are easy to manipulate—if a developer has access to the evaluations, it is easy for them to “game” the benchmarks, either by training on the evaluation itself (having the model memorize

the answers) or through more subtle means. This means that model evaluations often need to be kept private.

However, while the model evaluations themselves are private, the benefits they provide are public goods. Therefore, we need evaluation bodies whose results are public but whose methods remain private. While there is a market demand for such bodies, it is not entirely obvious that market mechanisms alone will produce them. They may well do so for some industries and use cases, but under-provide them for others. Thus, President Trump's AI Action Plan should leverage policy mechanisms to incentivize the creation of competitive markets for model evaluation providers – using the law to create the conditions for an AI evaluations ecosystem to flourish.

3. Foster Transparency: While major risks from AI may well present themselves in the near future, for the time being they remain speculative. Therefore, the role of public policy at this stage should be to foster greater insight into the nature of these risks and how frontier AI firms plan to mitigate them, rather than forcing any specific development or deployment practices on frontier AI companies.

The AI industry has already coalesced around a standard for such disclosures, all of which describe each respective company's approach to risk management for models with capabilities that could present legitimate potential for catastrophic risk. None of these documents address topics such as misinformation, bias, or other risks that could negatively impact freedom of expression. Their focus instead is on the most severe risks, which all companies developing AI acknowledge are genuine possibilities.

Making these documents public allows researchers, policymakers, and any other interested parties to understand how frontier AI companies are planning for major increases in model capabilities. This knowledge is a valuable public good.

Currently, companies release these documents as part of voluntary commitments. Given their utility, it would be prudent for a law to mandate that frontier AI companies continue releasing them to the public.

4. Create a Unified National Legal Framework: America's adoption of AI, and broader economic competitiveness, is being threatened by a patchwork of proposed and enacted state regulations. In the 2023-24 legislative session, states introduced more than 600 AI-related bills and passed more than 200¹. Many of these bills are anodyne or deal with aspects of law that most would agree are best left to state governments – for example, creating civil or criminal penalties for the knowing distribution of malicious deepfakes². Some, however, have far-reaching implications.

¹ Multistate.ai: [Artificial Intelligence \(AI\) Legislation](#), March, 2025.

² Dean Ball, The Deepfake Challenge: Targeted AI Policy Solutions for States (Oct. 2024), available at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5073572 (Accessed March 14, 2025)

AI will benefit from a clear and unified nationwide legal standard. If models with the potential to cause physical harm or loss of property are developed, this legal standard could be expanded with provisions affecting the following policies:

- I. An acknowledgement that frontier AI systems have such significant capabilities that there may be legally negligent means of developing and deploying such systems if their use results in physical harm or loss of property;
 - II. A means for developers to obtain safe harbor from the possibility of negligence liability for major harms if they adhere to certain development and deployment standards promulgated by either the federal government (e.g. the US AI Safety Institute) or a multi-stakeholder public-private standards body.
5. Invest in Longer-Term Governance: Given the speed with which AI capabilities can proliferate throughout the world it will likely be difficult or impossible for any government to maintain durable control over the development and use of very capable neural networks. However, just as the internet was fundamentally enabled by technical protocols, AI adoption may also require new protocols. We recommend considering the following suggested list of new protocols:
- I. Facilitate validation of personhood and identity in digital environments;
 - II. Reliably identify AI agents as such, and connect them back to the users on whose behalf they are acting; and
 - III. Ensure auditability of agent-agent interactions.

The private sector may create these protocols on their own, or solve these problems another way – but the area deserves further study in the AI Action Plan.

Accelerate AI Adoption in Government

AI presents governments with an immense opportunity to transform public service delivery. Far more efficient—and far better—government is within reach. To achieve this vision, we recommend that President Trump’s AI Action Plan include the following:

1. Revise the Office of Management and Budget’s M-24-10³ and M-24-18⁴ memos on agency use and procurement of AI to:
 - a. Narrow the definition of “safety-impacting AI” to focus exclusively on situations where physical harm is a possibility;
 - b. Narrow the definition of “rights-impacting AI” to focus on Constitutionally protected rights;
 - c. Eliminate requirements for agencies to consider the environmental impact of their AI use or of their AI system provider’s AI training/use; and

³ Executive Office of the President, Office of Management and Budget, Memorandum-24-10 (March 2024), available at <https://www.whitehouse.gov/wp-content/uploads/2024/03/M-24-10-Advancing-Governance-Innovation-and-Risk-Management-for-Agency-Use-of-Artificial-Intelligence.pdf>, (Accessed March 14, 2025)

⁴ Executive Office of the President, Office of Management and Budget, Memorandum-24-18 (Sept 2024), available at <https://bidenwhitehouse.archives.gov/wp-content/uploads/2024/10/M-24-18-AI-Acquisition-Memorandum.pdf> (Accessed March 14, 2025)

- d. Create a pathway for generalist AI systems (language models and agentic systems based upon language models) to be approved for a broad range of uses at either an agency of government-wide level, as opposed to creating a separate compliance process for each use case of such a system.
2. Modernize agency data government and retention/access policies to ensure that these policies do not present undue burdens to the adoption of AI;
3. Ensure sufficient construction of data centers meeting federal security requirements for both sensitive and classified data is taking place;
4. Revise the role of the agency “Chief AI Officer,” mandated by President Biden’s Executive Order on AI, to focus on driving adoption within agencies in addition to mitigating risks from AI; and
5. Eliminate the requirement in the 2023 Advancing American AI Act⁵ for agencies to publish inventories of their AI use cases, or make clear that general descriptions of some uses (for example, “using language models for back office process automation”) is sufficient, as opposed to an itemized inventory of each AI use.

Building, Permitting, and Semiconductor Manufacturing

AI is different from many other digital technologies in that it requires large-scale industrial infrastructure. Already, an industrial buildout of unprecedented speed and scale is taking place—by the end of the decade, it may be the largest such buildout in American history. In addition to data centers and the energy generation facilities needed to power them, AI will require cutting-edge semiconductor manufacturing capacity. Just like the data centers, the ideal outcome would be for many of those manufacturing facilities to be on American soil.

Accomplishing this will require large-scale reform to environmental regulation. America’s most successful environmental regulations rely on mandating objective standards for crucial things like air and water quality. These objective standards led to massive reductions in measurable pollution over the last several decades.

Permitting reform is the most important step—in some ways the most meaningful AI-related policy decisions the federal government can make in the short-term⁶. But it is far from the only step. In particular, ensuring that America retains a fundamental advantage in semiconductor manufacturing is essential.

In the longer term, it will also be crucial for American researchers to remain at the forefront of leading-edge semiconductor development. While it is true that TSMC has taken the lead in this regard, it remains the case that most of the fundamental advancements to semiconductor manufacturing were pioneered by Intel and other American firms. In recent years, it has become both more expensive and slower to move the frontier of semiconductors. New process nodes

⁵ Pub. L. No. 117-263 (May 2022), available at <https://www.congress.gov/bills/117/congress/house-bills/7776/text> (Accessed March 14, 2025)

⁶ Thomas Hochman and Aidan Mackenzie, How the White House Can Reform NEPA (Feb. 2025) available at <https://www.thefai.org/posts/how-the-white-house-can-reform-nepa> (Accessed March 14, 2025); Thomas Hochman, A Permitting Reform Wishlist for Congress (Feb. 2025) available at <https://www.greentape.pub/p/a-permitting-reform-wishlist-for> (Accessed March 14, 2025)

(shrinking the size of transistors to allow for more computationally performant and efficient chips) are taking closer to 36 months than 24, and the cost of achieving a new node has been rising for the past decade. As transistors become smaller, we are pushing up against fundamental physics in novel ways. All of this suggests that radically new approaches to every step in the semiconductor manufacturing process will be needed.

America is well-positioned to lead the way with these new approaches. Congress already allocated \$11 billion for the National Semiconductor Technology Center, a public-private research consortium that would do exactly this. If it is properly structured and led to pursue truly radical new research directions, it can become a world-leading research body, eventually paving the way to relying primarily or exclusively on private, rather than taxpayer, funding⁷.

Transforming Science

AI stands to massively improve the research productivity of scientists⁸. Today, a scientist at a research university might have a small team of graduate students and other collaborators at their disposal. Together, they navigate the laborious process of keeping up with the latest research, ideating potential experiments, conducting experiments, processing and analyzing data, writing papers—and of course, applying for grants. AI tools of all kinds—from narrow models designed to automate tasks to advanced generalist agents that can serve as a kind of “automated researcher”—promise to fundamentally change the means of producing science itself.

Many of the fundamental AI tools already exist or are on the development roadmaps of leading AI companies. But to take full advantage of these opportunities, science will need new infrastructure that gives scientists the ability to perform orders of magnitude more experiments than were previously possible. Government can support this infrastructure by investing in the creation of automated, “self-driving” labs. Many domains of science are not designed for industrial-scale experimentation, and as a result a great deal of scientific equipment is primarily built for the artisanal nature of science today. By creating demand for new kinds of lab equipment, built for massively scalable experimentation, government can foster a new wave of scientific progress.

To do this, the Department of Energy, the National Institute of Standards and Technology, the National Science Foundation, and other key federal agencies should:

1. Build cross-departmental science datasets and data infrastructure;
2. Partner with the private sector to develop science foundation models or, where sensitive data is being employed, build such models within government; and
3. Create a competitive bidding process or “grand challenge” for private firms to offer proposals on the creation of robotic labs (also referred to as “self-driving labs”).

⁷ Arrian Ebrahimi and Jordan Schneider, How to Make the NSTC A Moonshot Success, (April 2024) *available at* <https://ifp.org/how-to-make-the-nstc-a-moonshot-success/> (Accessed on March 14, 2025)

⁸ Dean Ball, Accelerating Materials Science With AI And Robotics (Nov. 2024) *available at* <https://fas.org/publication/accelerating-materials-science-with-ai-and-robotics/> (Accessed on March 14, 2025)

Ensuring AI Benefits Workers

The impact of AI on the labor market is uncertain. While many economists believe it will primarily augment human workers, others are concerned that there could be near-term shocks in specific sectors. The President should direct the data collection agencies of the federal government – the Bureau of Labor Statistics, the Census Bureau, the Bureau of Economic Analysis, etc. – to begin collecting data relevant to understanding the role of AI in the economy. This could include measurements of human task content, work process reorganization, human-AI collaboration metrics, skill demand dynamics, augmentation versus substitution effects, and more, collected via employer surveys and other means.

Building Genuine AI Security

AI security is necessary to enable widespread adoption of frontier technology, and another area where President Trump's AI Action Plan can lead.

If, for example, large-scale generalist AI systems develop capabilities with military applications, the Department of Defense (DoD) will want highly secure infrastructure on which to deploy these systems. No AI data center in America—and likely the world—has security characteristics sufficient to withstand a forceful cyberattack by a nation-state actor. But the DoD can lead the way by building a data center that meets the highest standards of cybersecurity, sufficient for the storage and analysis of Top Secret and Sensitive Compartmentalized Information. Such a facility would likely require both hardware and software innovations and is unlikely to be necessary for commercial use—making it an ideal target for government investment. However, some innovations made in the process could trickle down to the private sector, as has happened frequently in the history of American defense innovation.

Another field that could potentially benefit from defense innovation is AI alignment and control. While existing, private sector alignment methods may be sufficient for consumer and business applications, military uses of AI systems—where lives are on the line—may require more rigorous control and safety mechanisms. For example, autonomous systems with lethal capabilities may require mathematically verifiable guarantees⁹ that they will not engage in friendly fire. Again, guarantees of this kind may not be necessary for most non-military use cases, but just as with the data center innovations mentioned above, this research could eventually make its way into broader business applications (say, in medicine).

Finally, these recommendations would be incomplete without mentioning the criticality of export controls on advanced AI compute and semiconductor manufacturing equipment. Other plans submitted as part of this RFI are likely to go into great detail on this topic, but here we will mention just one area of potential reform: the Biden Diffusion Framework¹⁰.

⁹ David Dalrymple et al., Towards Guaranteed Safe AI: A Framework for Ensuring Robust and Reliable AI Systems, (May 2024) available at [arXiv:2405.06624](https://arxiv.org/abs/2405.06624) (Accessed March 14, 2025)

¹⁰ U.S. Department of Commerce, Bureau of Industry and Security, Framework for Artificial Intelligence Diffusion (Jan. 2025) Available at <https://www.federalregister.gov/documents/2025/01/15/2025-00636/framework-for-artificial-intelligence-diffusion> (Accessed on March 14)

The Diffusion Framework purports to regulate the distribution of AI hardware throughout the world. On top of this, however, it also regulates the export of closed-source model weights. It uses a flop-based threshold, which means that large pretrained models (like xAI's Grok 3) are subject to far heavier regulations than lighter-weight but equally capable models (such as OpenAI's o3-mini, particularly when configured in "high reasoning" mode). Moreover, these regulations are onerous, requiring data centers in other countries serving consumer chatbots to meet FedRamp High and, potentially, other security standards. The model weight component of the Diffusion Rule should be relaxed or eliminated. Relaxing country GPU caps on some Tier 2 countries may also be a potent diplomatic tool.

CONCLUSION

The American people are ultimately positive, hopeful, and pragmatic. They want to believe in a hopeful future for themselves and their families. They want AI to transform society for the better, and they want common-sense decisions that keep them safe and improve their day-to-day lives along the way. This societal-driven vision is a future where human-controlled AI systems enhance rather than replace human intelligence, creating a world with greater freedom for individuals, expanded possibilities for humanity's collective advancement, and widespread societal benefits that reach all communities. By maintaining our commitment to democratic principles, free enterprise, and human dignity, we can ensure that AI serves as a force for liberation and flourishing rather than constraint and displacement.

By approaching this challenge with wisdom, humility, openness, and a clear commitment to American values, we can navigate the AI revolution not as passive observers but as active architects of a more prosperous, just, and free society. The moment for reinvention is now. We must seize it together.

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